

```
1  #include<stdio.h>
2  #include<conio.h>
3
4  int main() {
5      int i, NOP, sum=0, count=0, y, quant, wt=0, tat=0, at[10], bt[10], temp[10];
6      float avg_wt, avg_tat;
7
8      printf("Total number of process in the system: ");
9      scanf("%d", &NOP);
10     y = NOP;
11
12     for(i=0; i<NOP; i++) {
13         printf("\nEnter the Arrival and Burst time of the Process[%d]\n", i+1);
14         printf("Arrival time is: \t");
15         scanf("%d", &at[i]);
16         printf("Burst time is: \t");
17         scanf("%d", &bt[i]);
18         temp[i] = bt[i];
19     }
20
21     printf("Enter the Time Quantum for the process: \t");
22     scanf("%d", &quant);
23
24     printf("\nProcess No \t\t Burst Time \t\t TAT \t\t Waiting Time\n");
25
26     for(sum=0, i=0; y!=0; ) {
27         if(temp[i] <= quant && temp[i] > 0) {
28             sum = sum + temp[i];
29             temp[i] = 0;
```

Total number of process in the system: 3

Enter the Arrival and Burst time of the Process[1]

Arrival time is: 0

Burst time is: 5

Enter the Arrival and Burst time of the Process[2]

Arrival time is: 1

Burst time is: 3

Enter the Arrival and Burst time of the Process[3]

Arrival time is: 2

Burst time is: 8

Enter the Time Quantum for the process: 2

Process No	Burst Time	TAT
Waiting Time		
Process No[2]	3	7
4		
Process No[1]	5	13
8		
Process No[3]	8	14
6		

Average Turn Around Time: 11.333333

Average Waiting Time: 6.000000

=== Code Execution Successful ===

```
30     count = 1;
31 }
32 else if(temp[i] > 0) {
33     temp[i] = temp[i] - quant;
34     sum = sum + quant;
35 }
36
37 if(temp[i]==0 && count==1) {
38     y--;
39     printf("Process No[%d] \t\t %d \t\t\t %d \t\t %d\n", i+1, bt[i], sum-
at[i], sum-at[i]-bt[i]);
40     wt = wt + sum - at[i] - bt[i];
41     tat = tat + sum - at[i];
42     count = 0;
43 }
44
45 if(i == NOP-1) {
46     i = 0;
47 }
48 else if(at[i+1] <= sum) {
49     i++;
50 }
51 else {
52     i = 0;
53 }
54 }
55
56 avg_wt = wt * 1.0 / NOP;
57 avg_tat = tat * 1.0 / NOP;
58
59 printf("\nAverage Turn Around Time: \t%f", avg_tat);
60 printf("\nAverage Waiting Time: \t%f", avg_wt);
```

```
61     getch();  
62     return 0;  
63 }
```